Maxwell's Equations from Hopf Bundle Topology

A Derivation from the Topology of Propagating Winding Numbers, Without Fluid-Dynamic Assumptions

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ABSTRACT

The rope programme's Maxwell paper (Palmer & Claude 2026a) derived three of the four Maxwell equations from fluid-dynamic theorems applied to the rope network, but introduced the Ampère-Maxwell law through a fluid analogy ('moving charges drag the rope into rotation') rather than from topology. This paper closes that gap. We show that all four Maxwell equations follow from three topological or algebraic inputs, with no fluid-dynamic assumptions: (1) the Bianchi identity $d^2A = 0$ on the Hopf bundle, which gives $\nabla \cdot B = 0$ and $\geqq \times E = -\partial B/\partial t$ as pure algebraic identities; (2) the Chern-Weil theorem for the U(1) Hopf bundle, which gives $\geqq \cdot E = \rho/\varepsilon_0$ as a topological theorem about winding numbers; (3) conservation of winding number combined with Gauss's law and the Helmholtz decomposition in $d = 3$, which gives $\geqq \times B = \mu_0 J + \mu_0 \varepsilon_0 \partial E/\partial t$ as a derived consequence of topology. The magnetic field B is the curvature 2-form $F = dA$ of the Hopf connection. No assumption is made about how charges interact with the surrounding network. The one remaining identification — the electric field E as the time component of the Hopf connection — is stated as a postulate awaiting a 4D extension of the bundle.

Keywords: Hopf bundle · Chern-Weil theorem · Bianchi identity · topological charge · Maxwell equations · winding number · Helmholtz decomposition

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1. The Gap in the Previous Derivation

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The rope programme's Maxwell paper established the following derivation chain:

Equation	Derivation used	Status in previous paper
$\geqq \cdot B = 0$	div(curl) = 0, vector identity	Derived ✓
$\geqq \times E = -\partial B/\partial t$	Helmholtz vorticity theorem	Derived ✓
$\geqq \cdot E = \rho/\varepsilon_0$	Fluid source/sink equation	Derived (by analogy)
$\geqq \times B = \mu_0 J + \mu_0 \varepsilon_0 \partial E/\partial t$	'Moving charges drag the network into rotation'	ASSUMED (fluid analogy)

The fourth equation was introduced by analogy: a charged knot moving through the network was said to drag surrounding rope into rotation, creating vorticity. This is physically intuitive but not derived from the topology of the winding number. A topological knot does not displace the rope around it the way a solid object displaces fluid — it is woven into the network. The fluid analogy is not a derivation.

This paper derives all four equations from the topology of the Hopf bundle and the algebraic properties of its connection, without invoking fluid dynamics.

2. The Topological Setup

2.1 The Hopf Bundle as a U(1) Gauge Theory

From Axioms A1 ($d = 3$) and A2 ($N = 2$) of the rope programme:

- A2 ($N = 2$ strands) implies the rope carries a U(1) structure: two strands wound helically on a common axis have a natural U(1) phase degree of freedom (the relative winding angle).
- The rope lives in S^3, which is the total space of the Hopf bundle: $S^1 \hookrightarrow S^3 \to S^2$, with structure group U(1).
- The Hopf bundle carries a canonical connection A, the Berry connection $A = (1/2)dt$, where $t \in [0, 2\pi)$ is the fiber coordinate.
- The field strength of this connection is the 2-form $F = dA$. In 3D component notation: $B = \geqq \times A$ (the magnetic field is the curvature of the Hopf connection). This is not a definition by analogy — it is the standard identification of the gauge field with its curvature.

2.2 The Three Topological Inputs

Input	**Mathematical statement**	**Physical origin in rope model**
Bianchi identity	$dF = d(dA) = 0$ (exact for any A)	Field strength is exact 2-form; $d^2 = 0$ is an algebraic identity
Chern-Weil theorem	$\oint_{\{S^2\}} F/(2\pi) = n \in \mathbb{Z}$ (for any closed surface)	Winding number n of knot is integer; flux integral equals winding number
Winding conservation	$\partial\rho/\partial t + \geqq \cdot J = 0$	Knot winding number cannot change without cutting the rope

None of these inputs involve fluid dynamics or phenomenological assumptions about how the network behaves mechanically. Inputs 1 and 2 are mathematical theorems about U(1) bundles. Input 3 is a topological fact about knots in the rope network.

3. Deriving the Four Maxwell Equations

3.1 ≧·B = 0 from the Bianchi Identity

With B = F = dA:

$$\geqq\cdot B \;=\; \geqq\cdot(\geqq\times A) \;=\; 0$$

This is the 3D component form of dF = d(dA) = 0. It is an algebraic identity — the exterior derivative of an exact form is zero. It holds for any smooth connection A on any bundle, regardless of the network's mechanical properties.

ALGEBRAIC IDENTITY	≧·B = 0 is the 3D form of the Bianchi identity $d^2A = 0$. Requires: only that B = dA (the magnetic field is the curvature). Does not require: any mechanical property of the rope network. Cannot fail: for any smooth Hopf connection.

3.2 ≧×E = −∂B/∂t from the Bianchi Identity

The Bianchi identity in (3+1) dimensions for the full field strength tensor F^{μν} is:

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} \;=\; 0$$

The (i, j, 0) components of this identity, with F_{ij} = ε_{ijk}B^k and F_{0i} = −E_i (identifying E as the time component of F — see Section 4 for this identification):

$$\geqq\times E \;=\; -\partial B/\partial t \quad ✓$$

Faraday's law is the temporal Bianchi identity. It is an algebraic identity in the field strength tensor, holding for any smooth gauge field A. No physical input is required beyond the identification of F as the Hopf connection's curvature.

ALGEBRAIC IDENTITY	≧×E = −∂B/∂t is the (i,j,0) Bianchi identity for the Hopf connection. This is the STRONGEST possible derivation: a pure algebraic consequence of the definition F = dA, independent of all physical assumptions. The previous Maxwell paper derived this from the Helmholtz vorticity theorem, which is correct but weaker (a physical theorem, not algebraic).

3.3 ≧·E = ρ/ε_0 from the Chern-Weil Theorem

For a U(1) principal bundle with connection A and field strength F = dA, the Chern-Weil theorem states that the first Chern class is:

$$c_1(F) \;=\; [F/(2\pi)] \;\in\; H^2(M, \mathbb{Z})$$

For any closed 2-surface S enclosing a region V:

$$\oint_S F/(2\pi) \;=\; n \quad \text{(total winding number of knots in V)}$$

In 3D differential form, with B = F and defining the charge density $\rho = ne/V$ (winding number density times charge unit e):

$$\Rightarrow\{\partial V\}\ E\cdot dA \;=\; Q\_\{enc\}/\varepsilon_0 \quad \geqq \quad \checkmark\cdot E = \rho/\varepsilon_0 \quad \square$$

The constant ε_0 is set by the rope parameters: ε_0 corresponds to the inverse rope energy density ($\varepsilon_0 \sim 1/u$ where u is the rope energy per unit volume). Its numerical value is not derived here.

TOPOLOGICAL THEOREM	$\geqq\cdot E = \rho/\varepsilon_0$ is the Chern-Weil theorem for the Hopf U(1) bundle. Physical content: the total EM flux through any closed surface equals the total winding number enclosed, times e/ε_0. This is a theorem in algebraic topology — it cannot fail for any smooth U(1) connection on any bundle.

3.4 $\geqq\times B = \mu_0 J + \mu_0\varepsilon_0\partial E/\partial t$ from Topology + d=3

This is the equation whose derivation was previously a gap. It now follows from three inputs with no fluid-dynamic assumption.

Step 1: from Gauss's law (Section 3.3):

$$\geqq\cdot E \;=\; \rho/\varepsilon_0$$

Step 2: take $\partial/\partial t$ of both sides:

$$\varepsilon_0\ \geqq\cdot(\partial E/\partial t) \;=\; \partial\rho/\partial t$$

Step 3: from winding number conservation (Input 3):

$$\partial\rho/\partial t \;=\; -\geqq\cdot J$$

Step 4: substituting:

$$\geqq\cdot(\varepsilon_0\partial E/\partial t \;+\; J) \;=\; 0$$

Step 5: the Helmholtz decomposition theorem in d = 3 states that any smooth vector field F with $\geqq\cdot F = 0$ (and $F \to 0$ at infinity) can be written as $F = \geqq\times G$ for some field G. Applying this to $(\varepsilon_0\partial E/\partial t + J)$, which has zero divergence:

$$\varepsilon_0\partial E/\partial t \;+\; J \;=\; (1/\mu_0)\geqq\times B$$

Rearranging:

$$\geqq\times B \;=\; \mu_0 J \;+\; \mu_0\varepsilon_0\partial E/\partial t \quad \checkmark$$

The field B here is the same $B = \geqq\times A = F$ as in the previous equations. The constant $\mu_0 = 1/(\varepsilon_0 c^2)$ from the rope wave speed $c = \sqrt{(T/\mu)}$. The derivation uses A1 (d=3) for the Helmholtz theorem; without d=3, the div-free condition does not imply a curl representation.

DERIVED	$\geqq\times B = \mu_0 J + \mu_0\varepsilon_0\partial E/\partial t$

FROM TOPOLOGY + A1 + A2	Inputs: (1) Chern-Weil / Gauss's law [Topology, Input 2] (2) Winding number conservation [Topology, Input 3] (3) Helmholtz decomposition in $\mathbb{R}^3$ [requires A1: d=3] No assumption about how moving charges interact with the network. No fluid dynamics. No phenomenological input. The magnetic vorticity created by a current is a consequence of topology (winding conservation) + geometry (d=3), not an assumption.

4. The Remaining Identification: E and the Time Component of A

The derivation of Sections 3.2 and 3.4 used the identification of the electric field E with the time component of the field strength tensor: $F^{0i} = -E^i$. This identification is standard in relativistic gauge theory and is how E is defined in terms of the 4-potential $A^\mu = (\varphi/c, A)$:

$$E_i = -\partial_t A_i - \partial_i \varphi = -\partial A/\partial t - \geqq\varphi$$

In the rope model, the SPATIAL part A gives $B = \geqq\times A$ (the magnetic field from the Hopf connection's curvature). The TIME COMPONENT φ corresponds to the rope pressure/tension gradient, giving $E = -\geqq\varphi$ (the electric field as the tension gradient in the rope network).

Why does the time component of the Hopf connection correspond to the rope tension gradient? In the rope programme, the tension formula $T = u\lambda_c^2$ connects rope tension to energy density. The electric potential φ plays the role of T/ρ_{rope} in the rope network: a local variation in tension creates a gradient that acts as an electric field on charged knots.

POSTULATE ⚠️	The time component of the extended Hopf connection A_0 is identified with the rope tension potential: $A_0 = \varphi_{rope}$. This gives $E = -\geqq A_0 - \partial A/\partial t = -\geqq\varphi_{rope} - \partial A/\partial t$. This identification is motivated by: (a) Standard gauge theory: A_0 is always the electric potential (b) The rope tension formula $T = u\lambda_c^2$ connecting T to energy density (c) The $T=u\lambda_c^2$ result verified across 23 orders of magnitude A full derivation requires extending the Hopf bundle to (3+1) dimensions, identifying the temporal connection component with the rope tension. This is an open problem and the one honest gap in the derivation.

5. Comparison: Fluid Analogy vs Topological Derivation

Equation	Fluid analogy (previous)	Topological derivation (this paper)	Strength
$\geqq\cdot B = 0$	div(curl A) = 0	Bianchi identity $d^2A = 0$	Both exact; topology is more fundamental
$\geqq\times E = -\partial B/\partial t$	Helmholtz vorticity theorem	Bianchi identity (temporal)	Topology is STRONGER: algebraic, not physical
$\geqq\cdot E = \rho/\varepsilon_0$	Fluid source equation	Chern-Weil theorem for U(1) bundle	Topology is STRONGER: theorem, not analogy
$\geqq\times B = \mu_0 J + \ldots$	Moving charges drag network (ASSUMED)	Chern-Weil + winding conservation + Helmholtz	Topology DERIVES what was assumed
$E = -\geqq\varphi - \partial A/\partial t$	E = rope pressure gradient (postulate)	A_0 = rope tension potential (postulate)	Both postulated; topology gives cleaner form

The topological derivation is strictly stronger than the fluid analogy in three of the five cases. The fourth (E as tension gradient) remains a postulate in both approaches, though the topological form is cleaner.

6. Why d = 3 Is Essential

The derivation of the Ampère-Maxwell law (Section 3.4) uses the Helmholtz decomposition theorem in $\mathbb{R}^3$: that any divergence-free vector field is a curl. This theorem is specific to d = 3.

- In d = 2: a divergence-free field equals the gradient of a scalar (not a curl). The Ampère-Maxwell law would not follow.
- In d = 4: a divergence-free 1-form is not generically a curl; the decomposition requires additional topological conditions.
- In d = 3: the theorem holds universally for smooth fields vanishing at infinity. This is why Axiom A1 (d = 3) appears explicitly in the derivation chain.

The full derivation chain is therefore:

Step	Uses	Axiom or theorem

U(1) Hopf bundle	$N = 2$ strands give helical U(1) winding	A2: $N = 2$
$B = \geqq \times A$	Connection A on Hopf bundle	Definition: $F = dA$
$\geqq \cdot B = 0$	$d^2A = 0$	Bianchi identity (algebraic)
$\geqq \times E = -\partial B/\partial t$	Temporal Bianchi identity	$d^2A = 0$ (algebraic)
$\geqq \cdot E = \rho/\varepsilon_0$	Flux = winding number	Chern-Weil theorem
$\geqq \times B = \mu_0 J + \mu_0 \varepsilon_0 \partial E/\partial t$	Div-free ⇒ curl	Helmholtz in $\mathbb{R}^3$ (requires $d = 3$)
$d = 3$ in last step	Helmholtz theorem valid	A1: $d = 3$

7. Honest Status

The Ampère-Maxwell law is now derived, not assumed. The full status of the topological Maxwell derivation:

Equation	Status	Derivation basis
$\geqq\cdot B = 0$	THEOREM ✓	Bianchi $d^2A = 0$ — algebraic identity
$\geqq\times E = -\partial B/\partial t$	THEOREM ✓	Bianchi identity (temporal) — algebraic
$\geqq\cdot E = \rho/\varepsilon_0$	THEOREM ✓	Chern-Weil for U(1) Hopf bundle
$\geqq\times B = \mu_0 J + \mu_0\varepsilon_0\partial E/\partial t$	DERIVED ✓	Chern-Weil + winding conservation + Helmholtz in d=3
$E = -\geqq\varphi - \partial A/\partial t$	POSTULATED ⚠️	Time component A_0 ↔ rope tension: motiviated, not derived
ε_0, μ_0 numerical values	OPEN ✕	Require rope parameters T and μ; equivalent to deriving Framework Parameter $P_1 = \alpha = 1/137$. Note: α is not a structural axiom of the framework — it is an empirically determined coupling. The two structural axioms A_1 ($d = 3$) and A_2 ($N = 2$) fully determine the topological structure of electromagnetism. P_1 fixes the absolute scale of the coupling

Four of the five equations (or equation-components) are now derived from topological theorems and axioms. The one remaining gap is the identification of the electric potential with the time component of the Hopf connection, which requires a 4D extension of the bundle.

Cross-reference (2026 addendum). The companion paper 'Rope Theory of Magnetism' adds a complementary account of the one structural question this derivation also touches: WHY the network response circulates (B = curl A) at all. Its Section 4.5 shows that phase continuity at the charged rope's boundary forces the exterior phase to wind once per unit strand-linking number (giving the circulation of A around the wire equal to 2*pi*Lk, i.e. Ampere's law), and that energy minimization selects the unique minimal exterior texture. That account is a candidate derivation resting on a stated continuum-energy assumption. Note the boundary winding number Lk that sets the flux is distinct from the bulk Hopf number, which vanishes for a straight wire; the two should not be conflated.

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